The underlying TCP connection MAY be long-lived or short-lived depending on the use case. Unlike MRP, which does not rely on an underlying connection, a secure session over TCP is unusable when its connection is broken or is closed. Nodes MAY choose to remove the secure session when the connection goes down. If the session is retained after the connection goes away, then the session SHALL be marked appropriately so that the underlying connection is re-established before the session can be used again. Moreover, the session resumption state MAY be retained to expedite session establishment when the connection is re-established with the corresponding peer.

4.15.2. TCP Connection Management

A TCP connection between a pair of nodes requires appropriate configurations and active management in order to meet the responsiveness demands of the application.

4.15.2.1. TCP Connection Configuration

The TCP connection configuration parameters MAY include the following:

1. `Connection Establishment Timeout`: A node MAY use this option to configure the amount of time that it waits for an initiated connection establishment attempt to succeed, before giving up and notifying the application.
2. `TCP Keep Alive Messages For Connection Liveness`: TCP Keep Alive messages MAY be used to maintain liveness for long-lived connections. They MAY be used at both the client and server to configure the liveness for each half of the connection. The configurable parameters SHALL be:
   a. *TCP_KEEP_ALIVE_TIME* : The interval between the last data packet sent and the first keep-alive probe.
   b. *TCP_KEEP_ALIVE_INTERVAL* : The interval between subsequent keep-alive probes.
   c. *TCP_KEEP_ALIVE_PROBES* : The number of unacknowledged probes to send before considering the connection dead and notifying the application.
3. `TCP User Timeout`: The TCP User Timeout option specifies the amount of time that transmitted data may remain unacknowledged before the TCP connection is forcibly closed. This option MAY be used by a node to limit the time the connection stays open when the data flow is not progressing.

The total `TCP Keep Alive Timeout` is given by:

```
TCP Keep Alive Timeout = TCP_KEEP_ALIVE_TIME + TCP_KEEP_ALIVE_INTERVAL *
TCP_KEEP_ALIVE_PROBES
```

4.15.2.2. TCP Connection Closures And Re-establishment

The TCP connection MAY be closed by either side of the connection. A node MAY proactively close its connection when instructed by the upper layer. It MAY also close the connection based on its current state.

For example, it MAY close its connection under the following circumstances:

1. The `TCP Keep Alive Timeout` expires on an idle connection.